Industrial Internship Report on

” IoT Temperature and Humidity Monitoring System using ESP32, MQTT and Node-RED”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was (Real-time IoT Temperature and Humidity Monitoring using ESP32, MQTT, Node-RED and DHT22) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

ABSTRACT

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship focused on developing an IoT based Temperature and Humidity Monitoring System using ESP32, MQTT protocol, Node-RED dashboard and DHT22 sensor.

The project was designed to monitor environmental conditions in real time and visualize data using a live dashboard with alert system.

This internship provided practical exposure to IoT systems, MQTT communication, Node-RED dashboard development and embedded systems concepts. It was a valuable learning experience that improved my technical and problem-solving skills.

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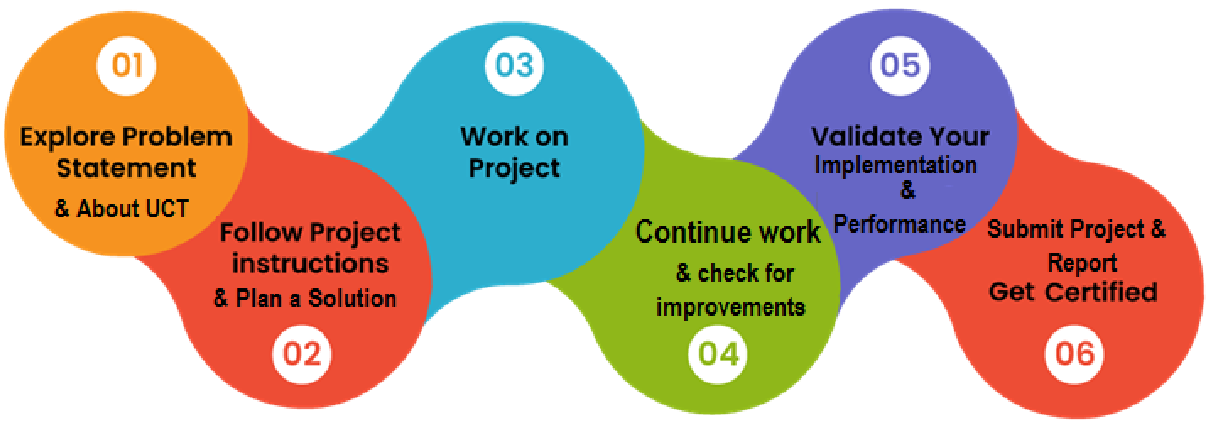
1 Preface

This internship provided an opportunity to learn Industrial IoT technologies and real-time monitoring systems. During the 6-week internship program, I learned concepts related to embedded systems, IoT communication protocols, MQTT, Node-RED dashboards and IoT platforms. The project assigned during the internship was an IoT Temperature and Humidity Monitoring System using ESP32 and DHT22 sensor. The system was designed to monitor environmental conditions and display real-time sensor data on a dashboard.



This internship improved my understanding of IoT systems and practical implementation of embedded technologies. I would like to thank upskill Campus, The IoT Academy and UniConverge Technologies Pvt Ltd for providing this valuable learning opportunity.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

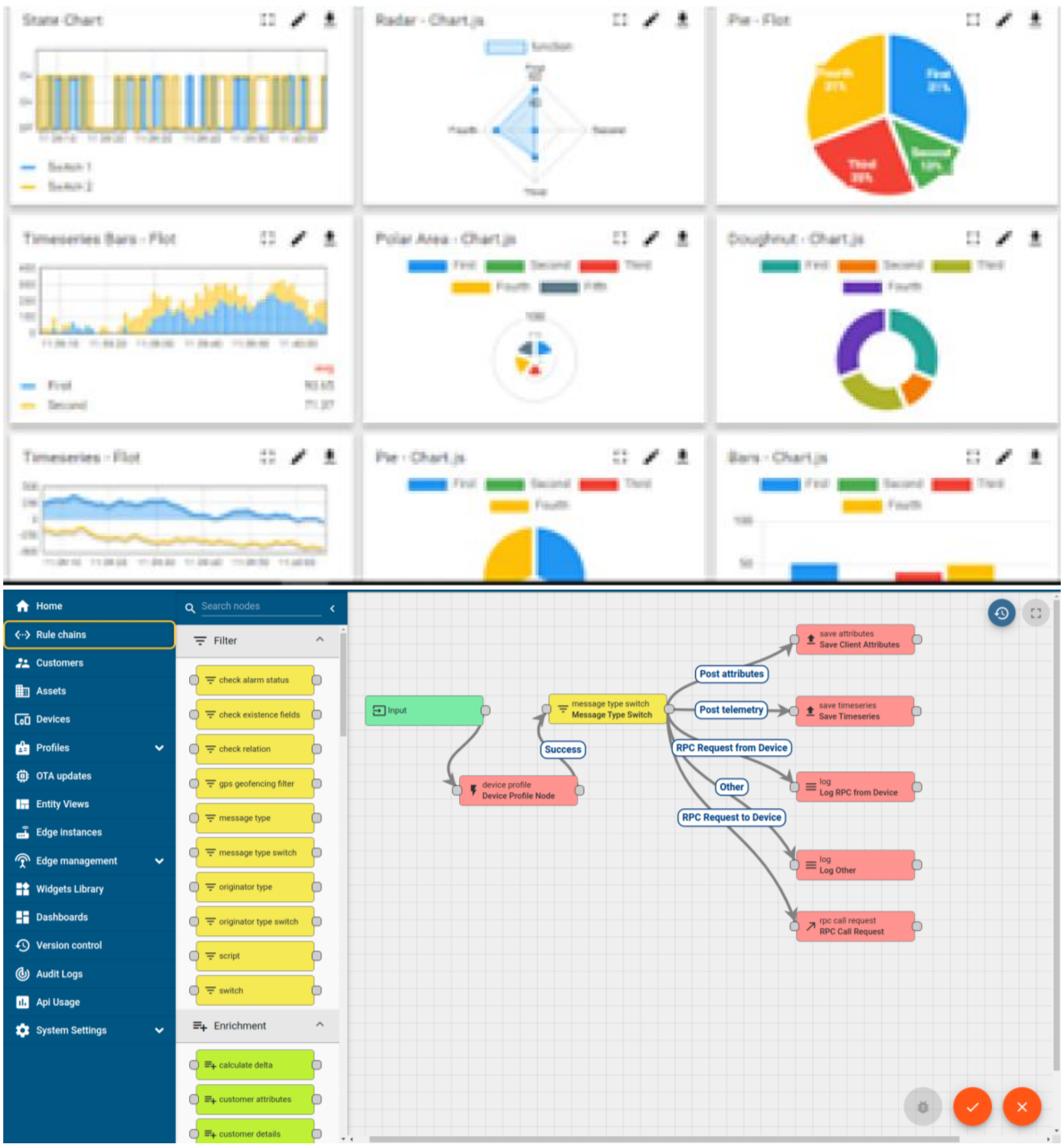
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

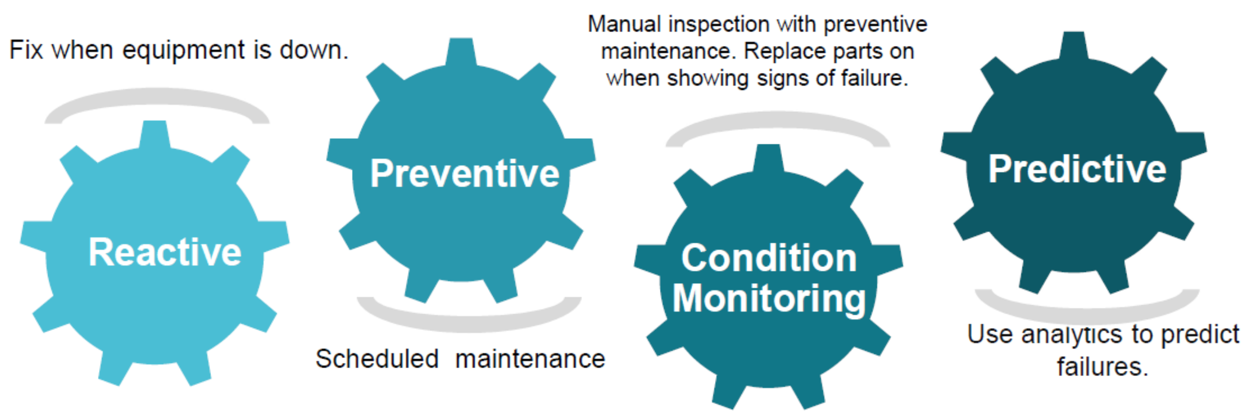
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

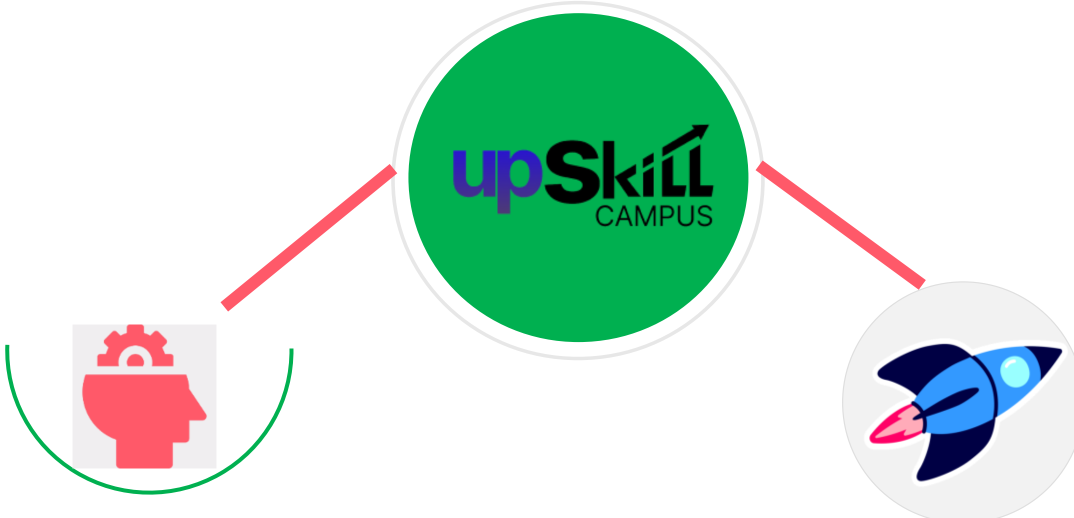
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

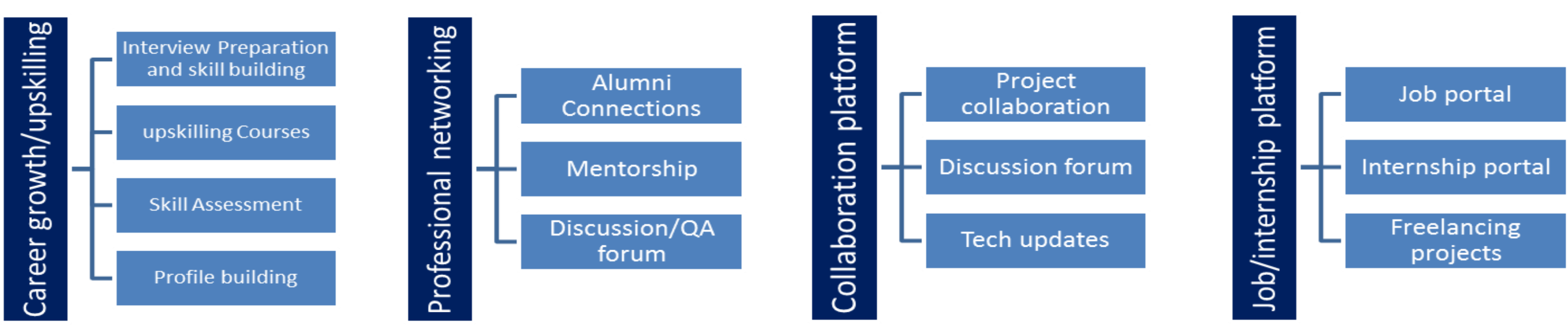
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Embedded Systems and IoT Study Material provided during Internship
[2] https://wokwi.com/
[3] https://mqtt.org/
[4] ESP32 Documentation
[5] Rise of Embedded Systems and IoT as Emerging Technologies 2023
[6] IoT Devices and Platforms Learning Video

2.6 Glossary

Terms	Acronym
Internet of Things	IoT
Message Queuing Telementry Transport	MQTT
Electronic Serial Peripheral Interface	ESP32
Digital Humidity and Temperature Sensor	DHT22
Graphical User Interface	GUI
HyperText Transfer Protocol	HTTP
Wireless Fidelity	Wi-Fi
Flow-based Programming Tool	Node-RED

3 Problem Statement

In the assigned problem statement

The objective of this project was to develop a real-time IoT based Temperature and Humidity Monitoring System using ESP32, DHT22 sensor, MQTT protocol and Node-RED dashboard.

The system monitors environmental conditions and displays live data through a graphical dashboard. Alert notifications are generated when temperature exceeds predefined limits.

4 Existing and Proposed solution

Traditional monitoring systems require manual checking and expensive industrial hardware systems. Such systems may not provide real-time remote monitoring capabilities

4.1 Code submission (Github link) : https://github.com/Arvind0506/upskillcampus.git

4.2 Report submission (Github link) : https://github.com/Arvind0506/upskillcampus.git

5 Proposed Design/ Model

The system consists of DHT22 sensor connected to ESP32. The ESP32 reads sensor values and sends data using MQTT protocol to Node-RED dashboard. The dashboard displays temperature and humidity values using gauges and graphs.

5.1 High Level Diagram

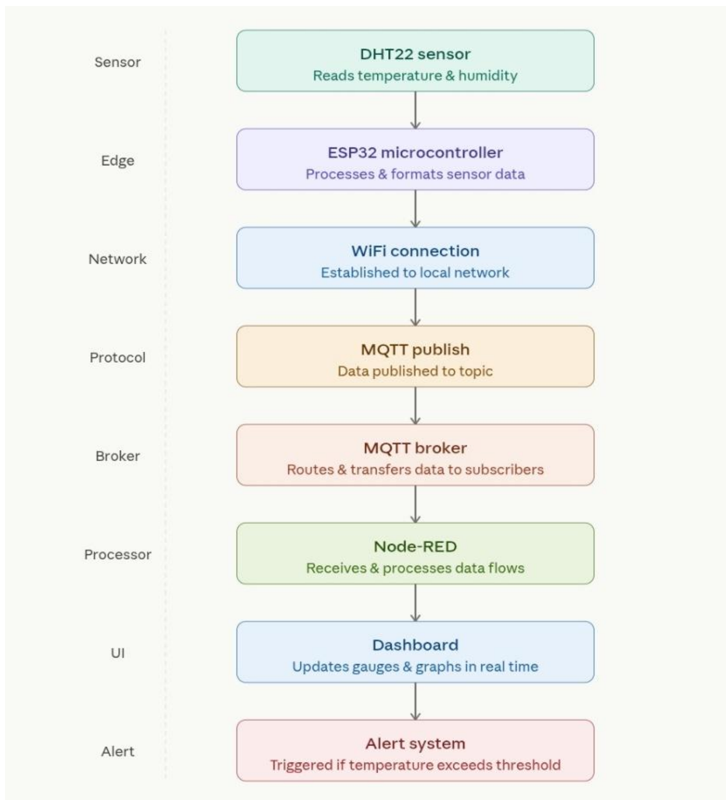


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

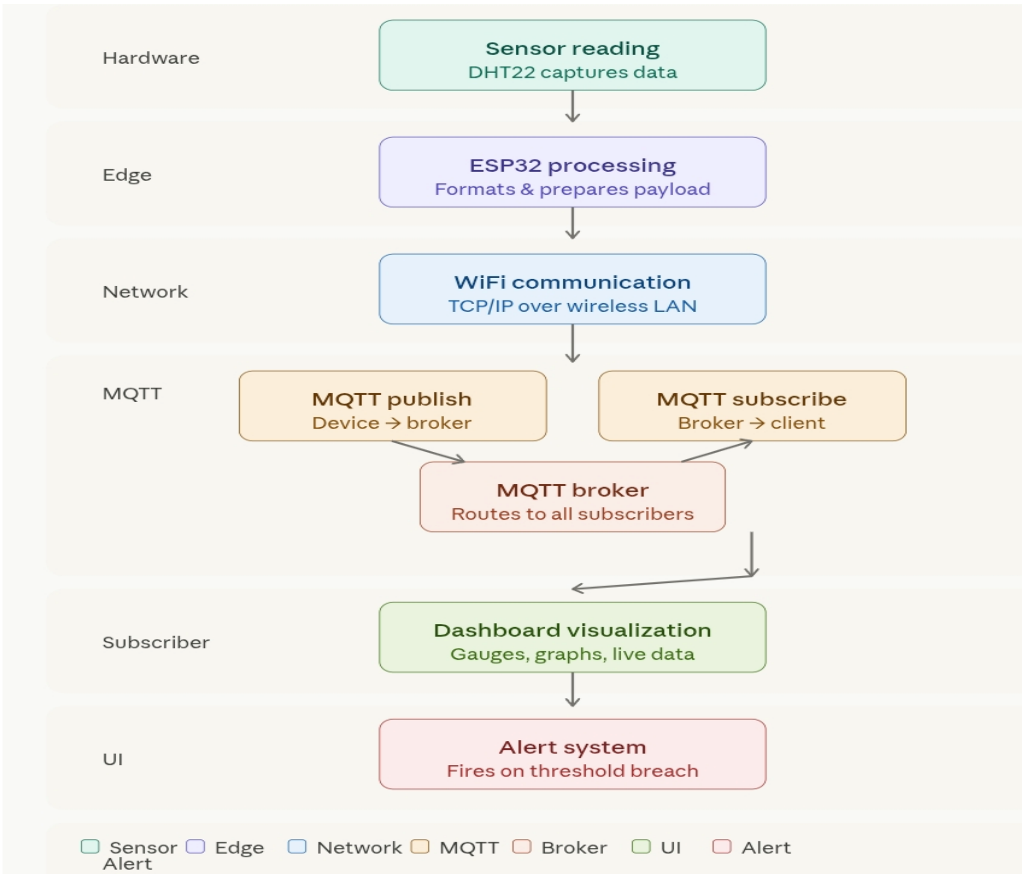


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The following interfaces were used in the proposed IoT Temperature and Humidity Monitoring System:

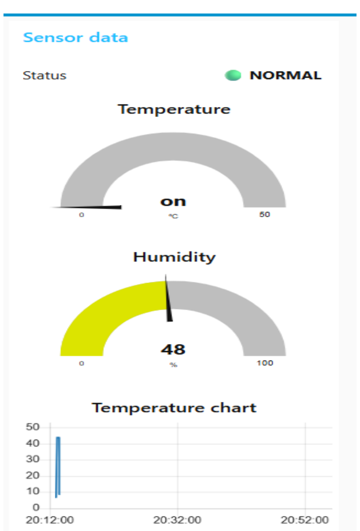
Sensor Interface:
DHT22 sensor interfaced with ESP32 for temperature and humidity data collection.

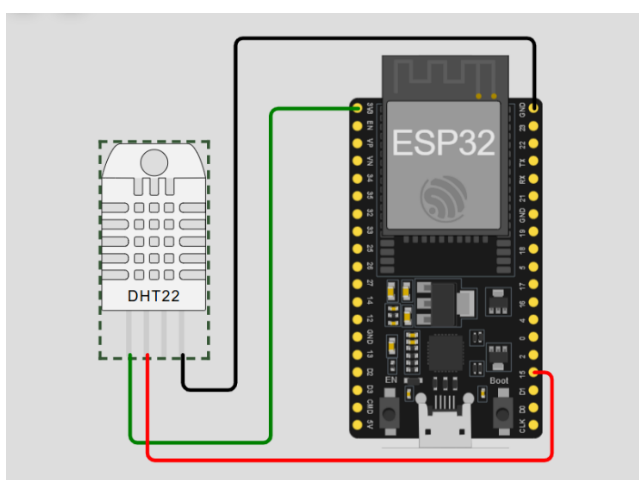
Communication Interface:
MQTT protocol was used for real-time data communication between ESP32 and Node-RED dashboard.

Network Interface:
WiFi network was used for internet connectivity and MQTT communication.

Dashboard Interface:
Node-RED dashboard was used for graphical visualization of sensor data using gauges, charts and alert notifications.

User Interface:
Users can monitor live temperature and humidity values through the dashboard interface.





6 Performance Test

The system was tested in Wokwi simulation environment. Sensor values were successfully transmitted using MQTT protocol and displayed on Node-RED dashboard in real time.

The dashboard successfully displayed temperature and humidity data along with alert notifications and graphical visualization.

6.1 Test Plan/ Test Cases

The IoT Temperature and Humidity Monitoring System was tested under different environmental conditions using Wokwi simulation and Node-RED dashboard.

Test cases included:

- Verification of temperature data transmission.
- Verification of humidity data transmission.
- MQTT broker connection testing.
- Real-time dashboard update testing.
- Alert generation testing for high temperature conditions.
- Graph visualization testing.

6.2 Test Procedure

1. ESP32 and DHT22 sensor were configured in Wokwi simulation.
2. Wi-Fi connection was established.
3. MQTT broker connection was verified.
4. Temperature and humidity values were published using MQTT protocol.
5. Node-RED dashboard subscribed to MQTT topics.
6. Live sensor values were displayed on gauges and charts.
7. Temperature values were manually changed in simulation to verify alert generation.
8. System performance and data visualization were monitored continuously.

6.3 Performance Outcome

The system successfully transmitted temperature and humidity data in real time using MQTT protocol.

The Node-RED dashboard displayed live sensor values, graphical charts and alert notifications correctly.

The system demonstrated stable communication between ESP32, MQTT broker and dashboard interface with smooth real-time monitoring performance.

7 My learnings

During this internship and project development, I learned the fundamentals of Embedded Systems and Internet of Things (IoT). I gained practical knowledge about ESP32 microcontroller, DHT22 sensor interfacing, MQTT communication protocol and Node-RED dashboard development.

I learned how real-time sensor data can be transmitted and visualized using IoT technologies. I also understood the working of MQTT broker communication and dashboard monitoring systems.

This project improved my programming, debugging and problem-solving skills. I also learned how to create industrial style IoT dashboards and how different IoT components communicate with each other.

Overall, this internship provided valuable practical exposure to IoT systems, embedded technologies and real-time monitoring applications, which will help me in future academic projects and career growth.

8 Future work scope

- Mobile application integration.
- Cloud database storage.
- Smart home automation integration.
- Remote monitoring through internet.
- AI based prediction and analysis system.